

Comparison of Four Non-Newtonian Blood Models in a 75% Stenosed Artery

1 Introduction

Blood flow in a severely stenosed artery is strongly affected by the rheological model used in the simulation. In this study, four non-Newtonian models were compared:

1. Cross model
2. Casson model
3. Carreau model
4. Power-Law model

The comparison was carried out using two hemodynamic quantities:

- Area-weighted average wall shear stress (WSS)
- Area-weighted average velocity magnitude

The objective was to determine how each model affects the predicted peak values and waveform trends in a 75 % stenosed artery under pulsatile flow conditions.

2 Expected Behavior Before Simulation

Before running the simulations, the expected behavior of the models was as follows:

- **Cross model:** Expected to predict relatively high wall shear stress because it can strongly reflect viscosity variation with shear rate.
- **Casson model:** Expected to produce moderate behavior due to inclusion of yield stress, which tends to resist motion at lower shear rates.
- **Carreau model:** Expected to provide smooth and realistic predictions, especially in arterial blood flow, because it is widely accepted for modeling blood rheology.

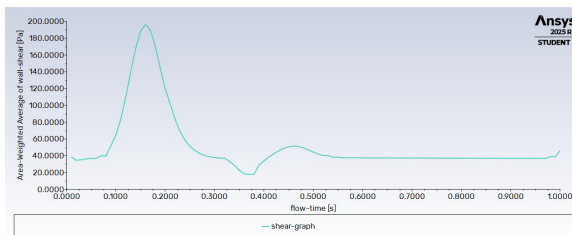
- **Power-Law model:** Expected to provide simplified but reasonable results, often close to Carreau in some flow conditions, although less accurate at very low shear rates.

From the physics of flow through a stenosis, it was also expected that:

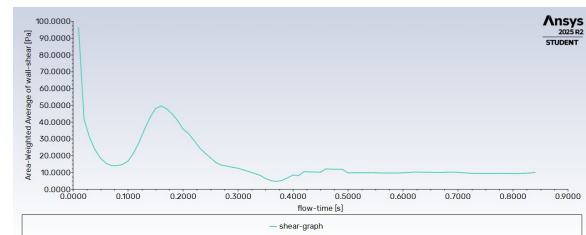
1. Velocity would increase sharply at the systolic phase due to acceleration through the narrowed throat.
2. Wall shear stress would rise strongly near the same period because of the increased velocity gradients at the wall.
3. The different rheological models would give different peak values, but the general waveform trend should remain similar.

3 Wall Shear Stress Results

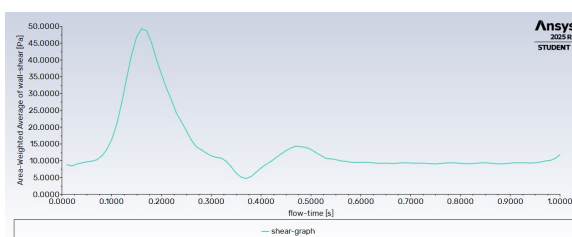
3.1 Individual WSS Plots



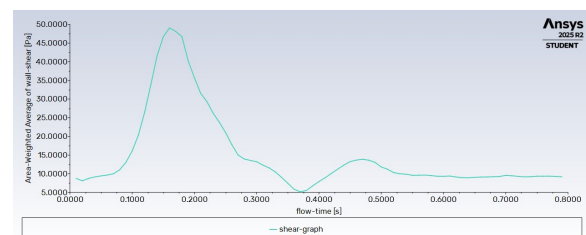
(a) Cross model: average wall shear stress



(b) Casson model: average wall shear stress



(c) Carreau model: average wall shear stress



(d) Power-Law model: average wall shear stress

Figure 1: Comparison of area-weighted average wall shear stress versus flow time for the four blood rheology models.

3.2 WSS Maximum Comparison

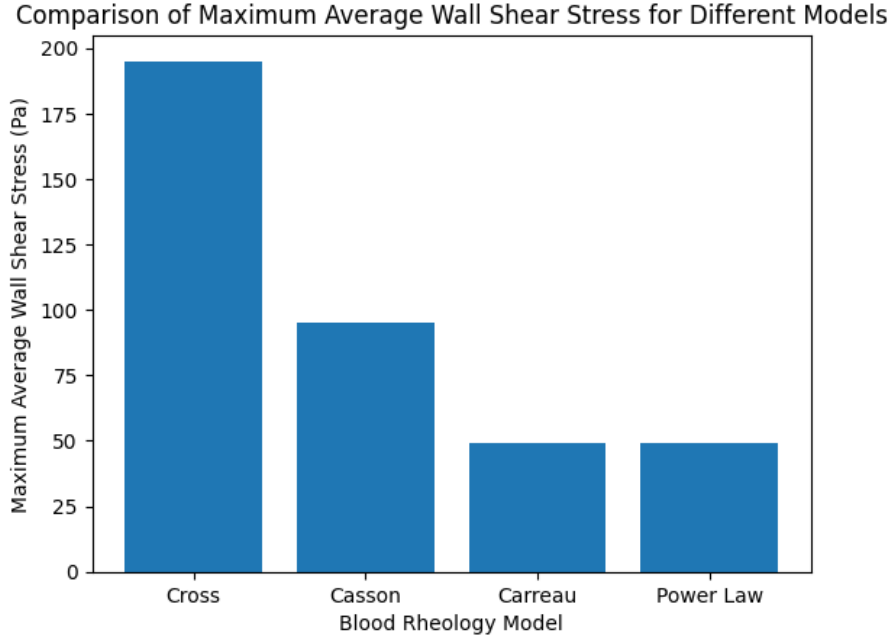


Figure 2: Bar chart comparing the maximum average wall shear stress predicted by the four models.

3.3 Discussion of WSS Behavior

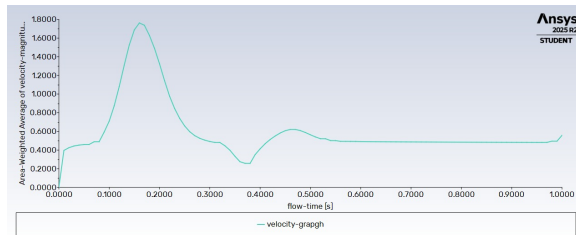
The wall shear stress results show that all four models follow the same overall pulsatile trend. In all cases, the wall shear stress rises sharply during the main acceleration phase of the cardiac cycle, reaches a peak, then decreases and approaches a lower quasi-stable level. This confirms the expected physical behavior in a severely stenosed artery, where the narrowing causes strong velocity gradients and therefore strong wall shear stress.

Among the four models, the **Cross model** produced the highest wall shear stress peak. This indicates that the Cross formulation was the most sensitive to the shear-rate changes occurring inside the 75 % stenosis. The **Casson model** produced a lower peak than the Cross model, which is consistent with the expected moderating effect of the yield-stress-based formulation. The **Carreau** and **Power-Law** models produced the lowest peak values and were relatively close to each other.

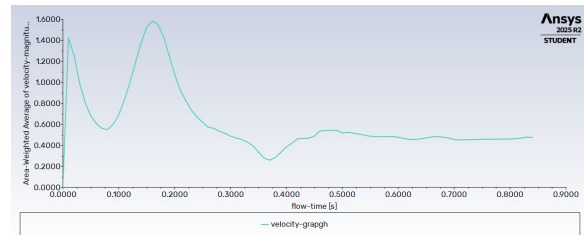
This behavior is physically reasonable. A severe stenosis is expected to cause large wall shear stress values, especially near peak systole. The simulations therefore confirmed the expected trend: the stenosis amplified shear stress strongly, and the choice of non-Newtonian model changed the magnitude of the peak without changing the overall waveform pattern.

4 Velocity Results

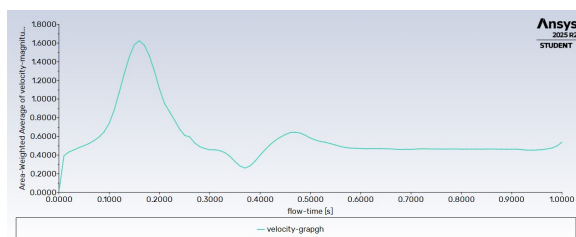
4.1 Individual Velocity Plots



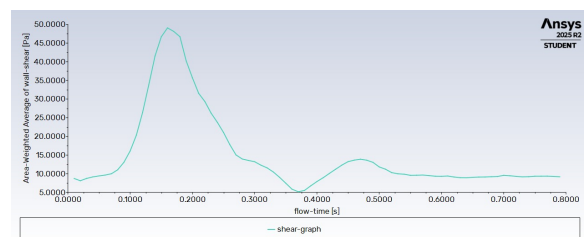
(a) Cross model: average velocity



(b) Casson model: average velocity



(c) Carreau model: average velocity



(d) Power-Law model: average velocity

Figure 3: Comparison of area-weighted average velocity magnitude versus flow time for the four blood rheology models.

4.2 Velocity Maximum Comparison

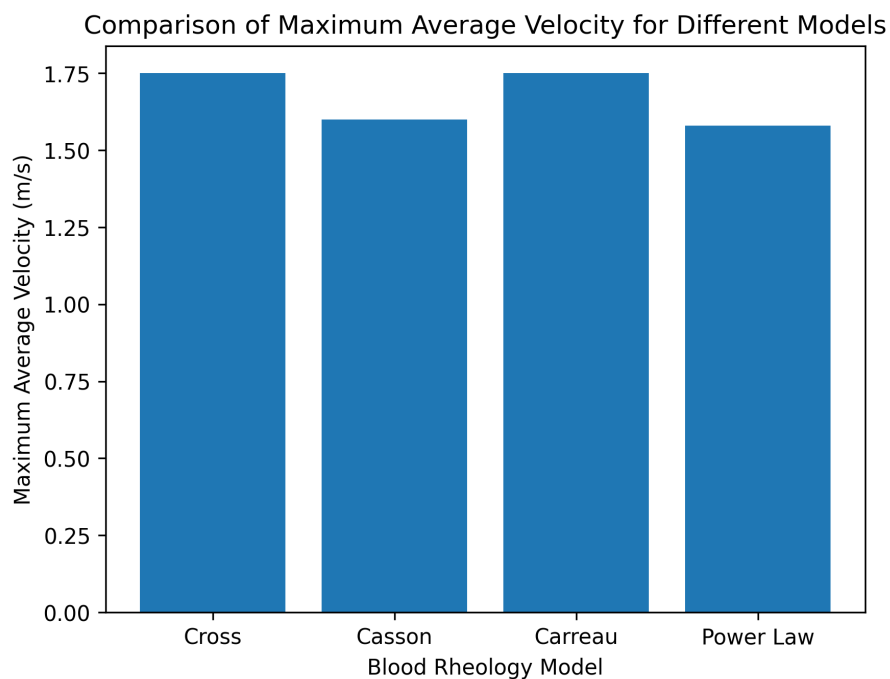


Figure 4: Bar chart comparing the maximum average velocity predicted by the four models.

4.3 Discussion of Velocity Behavior

The velocity plots also show a similar pulsatile waveform for all models. In each case, velocity increased rapidly during the main acceleration stage, reached a distinct peak, then decreased after the systolic phase. This is exactly what should happen in a stenosed artery, because the narrowing forces the fluid to accelerate through a reduced cross-sectional area.

The **Cross** and **Carreau** models predicted the highest maximum average velocity values, while the **Casson** and **Power-Law** models produced slightly lower peaks. Even though the differences in velocity were smaller than the differences observed in wall shear stress, the bar chart still shows that the rheological model has a measurable influence on the flow acceleration through the stenosis.

This result also agrees with expectations. Since stenosis causes a jet-like acceleration of flow, all models should predict a strong velocity peak. The simulation results confirmed this. At the same time, the rheological model changes the effective viscosity behavior, which explains the differences in the exact peak velocity values.

5 Direct Comparison Between the Four Models

5.1 Comparison in Terms of Wall Shear Stress

The four models can be compared as follows:

- **Cross model:** Highest wall shear stress peak; strongest sensitivity to the stenosis.
- **Casson model:** Intermediate wall shear stress prediction; lower than Cross but higher than Carreau and Power-Law.
- **Carreau model:** Lower and smoother wall shear stress prediction; widely considered reliable for blood flow in arteries.
- **Power-Law model:** Similar trend to Carreau, but based on a simpler formulation.

Thus, for wall shear stress, the models can be ranked approximately from highest to lowest response as:

$$\text{Cross} > \text{Casson} > \text{Carreau} \approx \text{Power-Law}$$

5.2 Comparison in Terms of Velocity

The velocity comparison shows a smaller separation between the models. However, the differences are still visible:

- **Cross and Carreau models:** Higher peak velocity values.

- **Casson and Power-Law models:** Slightly lower maximum velocity values.

Thus, for maximum average velocity, the models can be ranked approximately as:

$$\text{Cross} \approx \text{Carreau} > \text{Casson} \approx \text{Power-Law}$$

6 Did the Simulation Match the Expected Results?

Yes, the simulation results largely matched the expected physical behavior.

1. **Expected:** Severe stenosis should produce a strong velocity peak.
Observed: All four models showed a distinct acceleration peak in velocity.
2. **Expected:** Wall shear stress should increase strongly during peak flow due to high velocity gradients near the stenosis wall.
Observed: All four models showed a significant WSS rise during the same part of the cardiac cycle.
3. **Expected:** Different non-Newtonian models should change the magnitude of the response but not the overall waveform shape.
Observed: The waveform pattern remained similar for all models, but the peak magnitudes changed noticeably.
4. **Expected:** The Cross model could yield a stronger response, while Carreau should remain realistic and smooth.
Observed: The Cross model gave the highest WSS peak, while Carreau gave smoother and more moderate predictions.

Therefore, the simulations support the expected hemodynamic behavior of pulsatile blood flow through a 75 % stenosed artery and confirm that model selection affects quantitative predictions, especially the wall shear stress peak.

7 Conclusion

The four non-Newtonian blood models produced similar pulsatile waveform shapes but different peak magnitudes for both wall shear stress and velocity. The stenosis caused the expected acceleration of flow and corresponding increase in wall shear stress. Among the tested models, the Cross model predicted the most aggressive shear response, the Casson model gave intermediate behavior, and the Carreau and Power-Law models gave lower and smoother peak values.

These results indicate that the choice of blood rheology model is important when evaluating hemodynamic quantities in a stenosed artery. In particular, wall shear stress

is more sensitive to the rheological model than average velocity. The simulation therefore confirms both the expected effect of the stenosis and the expected influence of the non-Newtonian constitutive law.